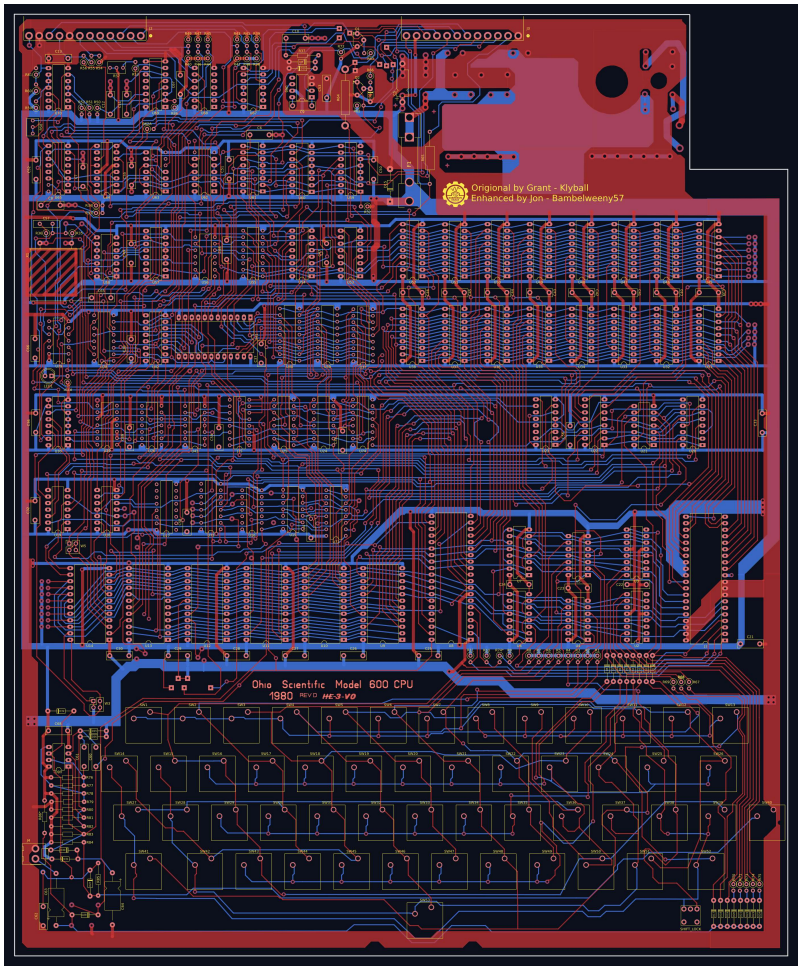
OHIO SCIENTIFIC

Superboard II

Model 600 Rev D – Cherry MX keyboard edition

Assembly and test guide



For board rev 1.0 (KiCad 10 project). Based on the Ohio Scientific Superboard Kit Assembly Manual, April 1982.
Original copper reconstruction: Grant – Klyball. Enhanced edition: Jon – Bambelweeny57, Submeson Brain Corporation.

Contents

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About this guide

This guide takes you through building the Superboard II Model 600 Rev D reproduction board with Cherry MX keyswitches, from a bare printed circuit board to a working computer that greets you with D/C/W/M ?.

It follows the order of Ohio Scientific's 1982 Superboard Kit Assembly Manual: low-profile parts first, then the keyboard, then the video section, and finally the processor, ROMs and RAM, with a test after each stage. Every step has been rewritten for this board. Part references, values, orientations and test points all come from the rev 1.0 KiCad design, so where this guide and the 1982 manual disagree, follow this guide.

What is different from the 1982 kit

- Keyboard: 54 Cherry MX keyswitch positions in place of the original keys, including a choice of two shift-lock positions.
- ROMs: the monitor (U13) and BASIC (U9–U12) sockets take 27C16 EPROMs or 2816 EEPROMs; the character generator (U41) is a 2716-type EPROM.
- Options: the original wire-jumper blocks are replaced by pin headers W3 and W5.
- Power: 5 V enters at the +5V_IN holes and passes through fuse F1 before the protection diode D15.
- Some reference designators were renumbered so the design works in KiCad (see Appendix E).

How to use this guide

- Read the whole guide before you start, as the original manual advises.
- Each step has a parts table and a board figure showing where the parts go. Tick the box beside each part as you fit it.
- Compass directions are used throughout. Hold the board component side up with the notched corner at the top right: the top edge, with the J2 and J3 connectors, is North and the keyboard is South.
- Do not skip the tests. Each one catches problems while they are still easy to fix.

Caution Never insert or remove chips while power is applied. Disconnect the supply before every IC step.

Tools and materials

Item	Notes
Temperature-controlled soldering iron	A fine conical or 1.6 mm chisel tip, set around 330–350 °C for leaded solder
Solder	0.5–0.7 mm rosin-cored solder; leaded 60/40 or 63/37 is easiest to work with
Flux pen, desoldering braid or pump	For cleaning up bridges
Side cutters and fine-nose pliers	For trimming and forming leads
Digital multimeter	Resistance, continuity, DC and AC voltage
Logic probe (optional)	Useful for the video and clock checks in Test 5
Isopropyl alcohol and a brush	For removing flux residue
EPROM programmer	To program the monitor, BASIC and character ROMs (see Section 9)

To use the finished computer

- A regulated 5 V DC supply rated at 3 A, regulation better than 4 % and ripple under 4

%.

- A composite video monitor, or a TV with a composite video input or an RF modulator.
- A cassette recorder for loading and saving programs (optional).
- A small speaker on J4 for the sound DAC (optional).

1 Getting to know the board and the parts

Figure 1-1 shows the board with every part position. The numbering follows the original: chips are U-numbers, resistors R, capacitors C, diodes D, and the keyswitches SW. The silkscreen on the board shows each reference, so you can always check a position against the figure.

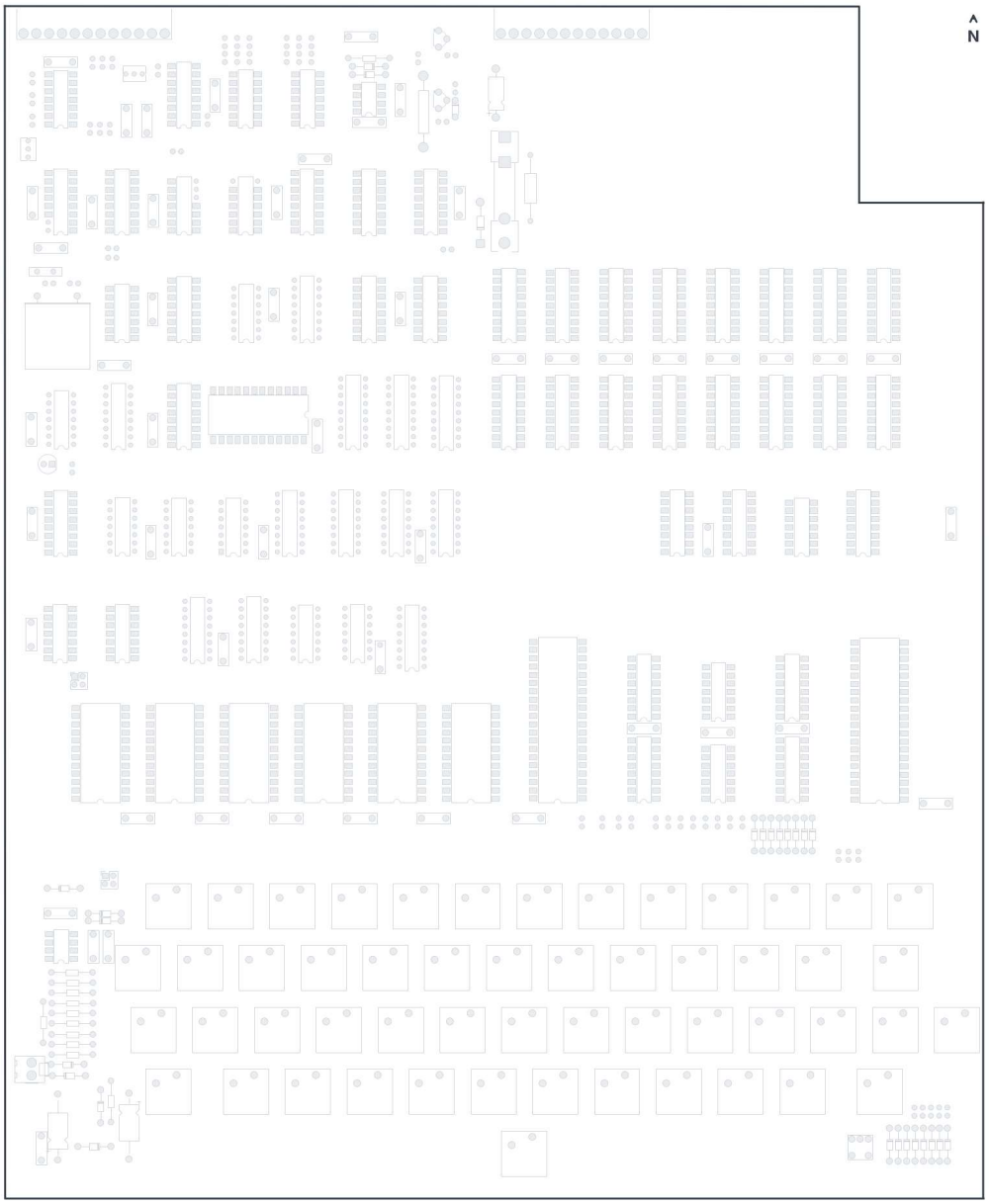


Figure 1-1 Board layout, component side, North at the top.

Board sections

Area	What is there
North-west	Oscillator, video timing counters, cassette interface (U29, U30, U57–U66, X1)
North centre	Serial (ACIA) and RS-232 parts, J2 and J3 edge connectors
North-east	Fuse F1, power entry and the program RAM (U31–U38, U45–U52)
Centre	6502 processor U8, monitor ROM U13, BASIC ROMs U9–U12, address decoding
West centre	Video RAM, character generator U41, video output

South	Keyboard switches SW1–SW54, keyboard decoding (U2–U5), sound DAC and reset circuit

Orientation rules

Part	How to orient it on this board
IC sockets and chips	Notch (pin 1 end) faces South on every chip except U41, whose notch faces East. The square pad is pin 1.
Diodes	The band (cathode) goes to the square pad and matches the band printed on the silkscreen. Each diode step says which way the band faces.
Electrolytic capacitors	The + lead goes in the square pad. C5 and C63 have + to the South; C64 has + to the North.
LED1	Flat side (cathode, shorter lead) faces East, into the square pad.
Transistors Q1, Q2	Match the flat face to the silkscreen outline.
Resistors	Not polarised. Fitting them all with the first colour band the same way makes checking easier.

Resistor colour codes

Value	Bands (4-band)	Value	Bands (4-band)
100 Ω	brown black brown	4.7 kΩ	yellow violet red
220 Ω	red red brown	8.2 kΩ	grey red red
390 Ω	orange white brown	10 kΩ	brown black orange
470 Ω	yellow violet brown	16 kΩ	brown blue orange
510 Ω	green brown brown	33 kΩ	orange orange orange
1 kΩ	brown black red	68 kΩ	blue grey orange
2 kΩ	red black red	100 kΩ	brown black yellow
3.9 kΩ	orange white red	1 MΩ / 10 MΩ	brown black green / brown black blue

The fourth band (gold, 5 %) is the tolerance. 1 % metal-film resistors have five bands; check them with your meter if in doubt.

Capacitor markings

Small film and ceramic capacitors are marked with a three-digit code: the first two digits are the value and the third is the number of zeros, in picofarads. So 104 = 100 000 pF = 0.1 µF, 103 = 0.01 µF, 102 = 0.001 µF, 151 = 150 pF, 680 or 68 = 68 pF and 270 or 27 = 27 pF.

> **Note** Do a dry run of each capacitor step: place all the parts loosely, check the values against the table, then solder.

2 Stage I – sockets and low-profile parts

Stage I fits everything that lies close to the board: IC sockets, diodes, resistors, capacitors and the smaller fittings. Work from the lowest parts to the tallest so the board sits flat while you solder.

Part 1 IC sockets

Use good-quality sockets throughout; turned-pin sockets are best. For each socket:

- Insert it with the notch facing South (U41 faces East).
- Turn the board over and bend two opposite corner pins to hold it.
- Check the position and orientation, then solder every pin. Solder the larger sockets first within each area.

Step 1 40-pin sockets

☐	Ref	Part	Notes
☐	U8	40-pin socket	6502 processor
☐	J1	40-pin socket	Expansion connector for a 610 board; fit it now if you may expand later

Step 2 24-pin sockets

☐	Ref	Part	Notes
☐	U9	24-pin socket	BASIC 1 ROM
☐	U10	24-pin socket	BASIC 2 ROM
☐	U11	24-pin socket	BASIC 3 ROM
☐	U12	24-pin socket	BASIC 4 ROM
☐	U13	24-pin socket	Monitor ROM
☐	U14	24-pin socket	6850 ACIA
☐	U41	24-pin socket	Character generator – notch faces East

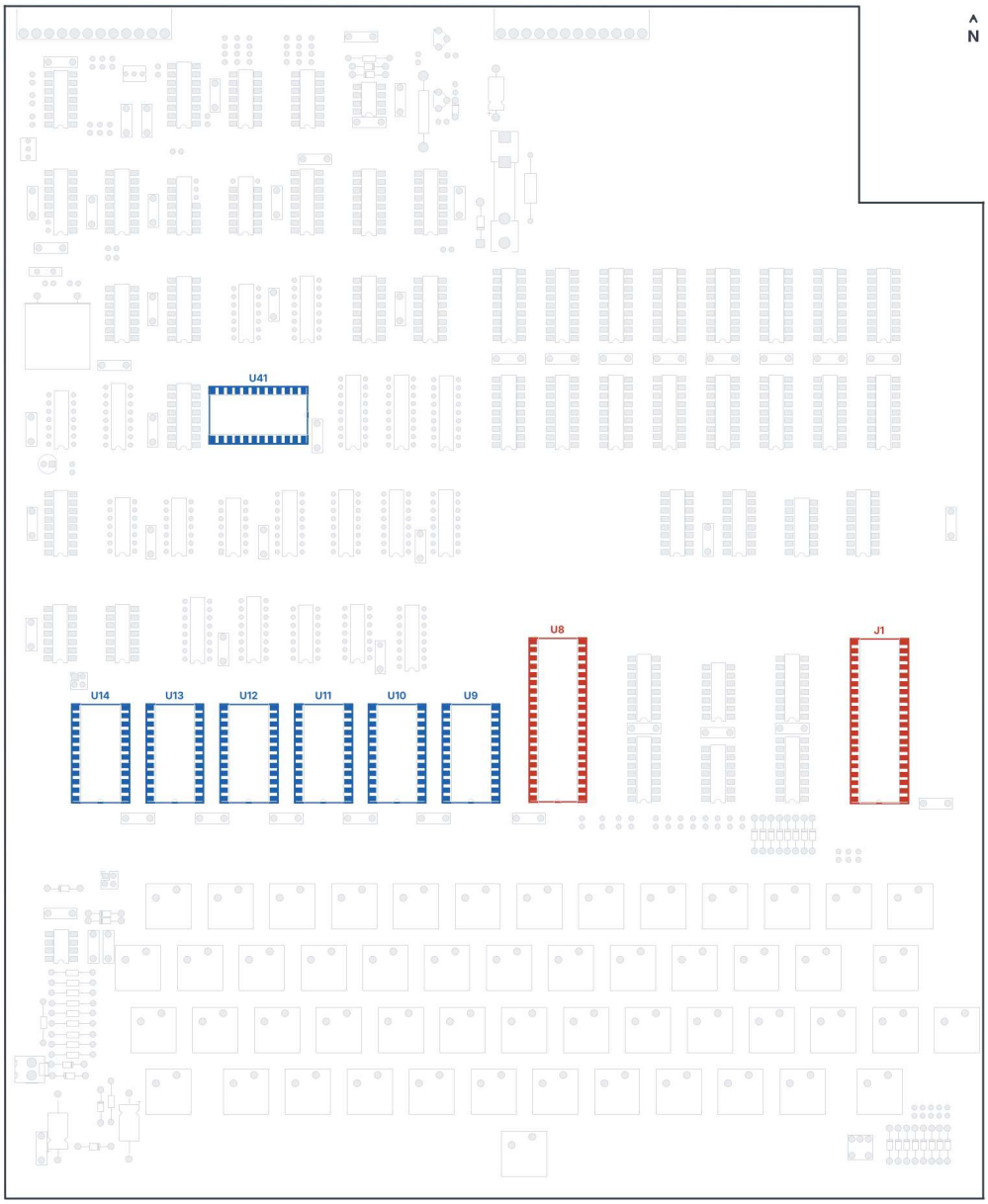


Figure 2-1 Steps 1 and 2: 40-pin sockets (red) and 24-pin sockets (blue).

Test 1 ROM socket wiring (after Step 2)

Purpose: check for solder bridges and confirm the ROM socket power pins. Use the resistance range of your meter on socket U9.

- Pins 12 and 18 should read as a short (both are ground: pin 18 is the ROM chip-enable, tied low for the 27C16).
- Pins 21 and 24 should read as a short (both are +5 V).
- Every other pair of pins should read open. Check pin 1 against pins 2–24, then pin 2 against pins 3–24, and so on.

Note The 1982 manual expects pins 18, 21 and 24 to be connected together. That was for the original 2316B ROM. On this board pin 18 goes to ground instead.

Step 3 18-pin sockets

Program RAM U31–U38 and U45–U52, video RAM U39 and U40, and colour RAM U77.

☐	Refs	Count	Notes
☐	U31–U38, U45–U52	16	2114 program RAM
☐	U39, U40	2	2114 video RAM
☐	U77	1	2114 colour RAM (optional, used with the colour option on J75)

Step 4 16-pin sockets

☐	Refs	Count	Notes
☐	U2, U3, U17, U20, U22, U23, U24, U25, U30, U42, U53, U54, U55, U57, U59, U60, U61, U64, U65, U69, U72, U73, U78	23	
☐	U6, U7	2	8T28 bus buffers to J1 – only needed with an expansion board
☐	U74, J75	2	Colour option (U74 8T28, J75 colour connector) – optional

Step 5 14-pin sockets

☐	Refs	Count	Notes
☐	U4, U5, U15, U16, U18, U19, U21, U29, U43, U56, U58, U62, U63, U70, U76, U79	16	
☐	U67, U68	2	External serial buffers on J2 – normally not fitted

Step 6 8-pin sockets

☐	Refs	Count	Notes
☐	U66, U80	2	CA3130 and LM393. The 1982 kit soldered these in directly; sockets make replacement easier.

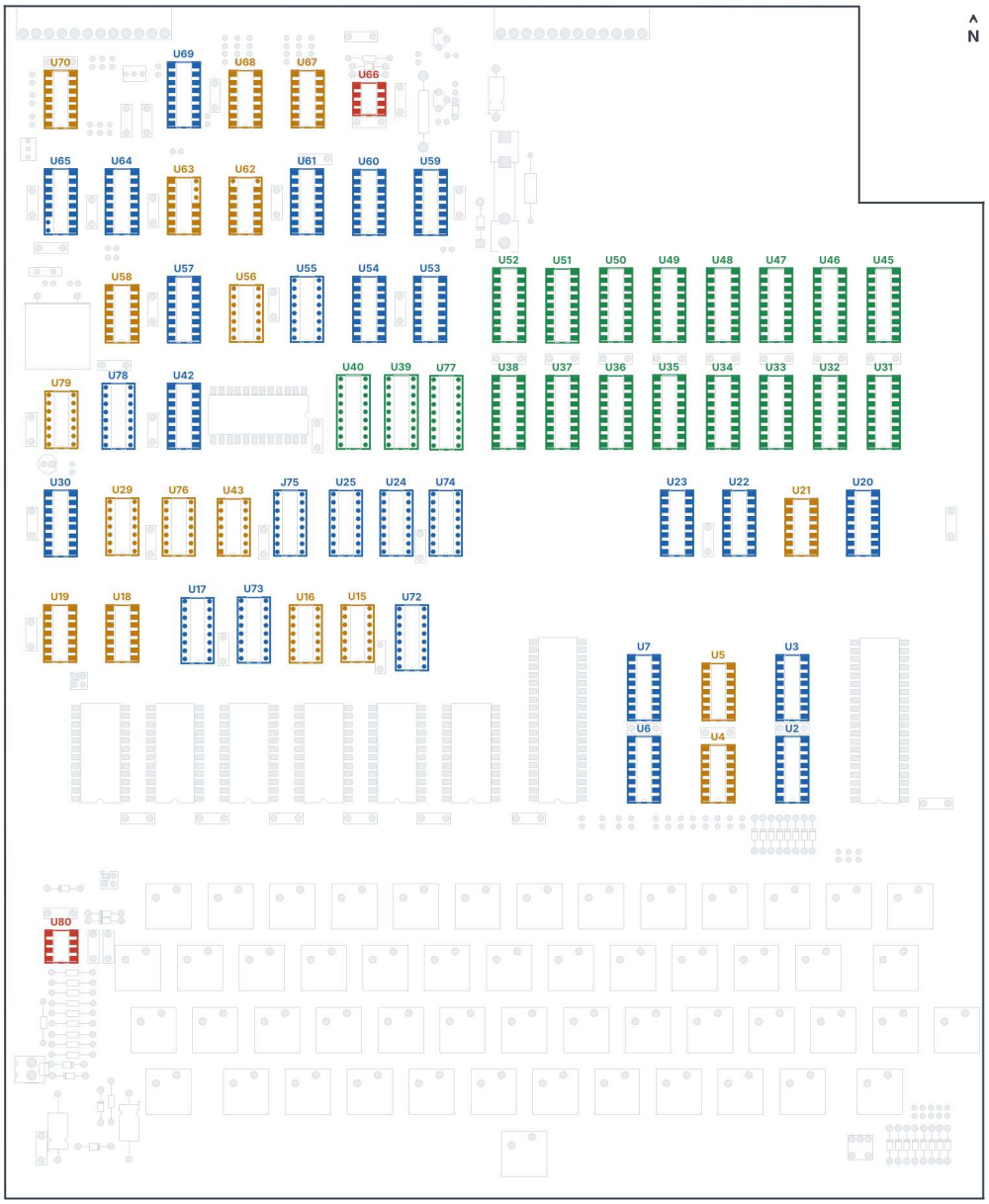


Figure 2-2 Steps 3–6: 18-pin (green), 16-pin (blue), 14-pin (orange) and 8-pin (red) sockets.

Test 2 Solder bridges (after Step 3)

On socket U31 check every pin against every other pin (1 to 2–18, 2 to 3–18 and so on). All should read open, including pin 9 (ground) to pin 18 (+5 V). Because U31 connects to the rest of the RAM and many other sockets, this also checks those for bridges.

Test 3 Address and data continuity (after Step 6)

Each line below should read as a short circuit between all the pins listed.

Line	U8 pin	U13 pin	U31 pin
A0	9	8	5
A1	10	7	6
A2	11	6	7
A3	12	5	4
A4	13	4	3
A5	14	3	2

A6	15	2	1
A7	16	1	17
A8	17	23	16
A9	18	22	15

Line	U8 pin	U13 pin	RAM pin
D0	33	9	U31-14
D1	32	10	U31-13
D2	31	11	U31-12
D3	30	13	U31-11
D4	29	14	U45-14
D5	28	15	U45-13
D6	27	16	U45-12
D7	26	17	U45-11

If a line reads open, look for an unsoldered pin on that socket, then check the track with the meter step by step.

Part 2 Diodes

There are 26 small-signal diodes (1N914 or 1N4148) and one rectifier (1N4001). Bend the leads to fit the hole spacing, insert with the band towards the square pad, solder and trim. 1N4148 is a direct replacement for 1N914.

Step 7 Keyboard row diodes

☐	Refs	Part	Band faces
☐	D1–D8	1N914	North

Step 8 Sound DAC diodes

☐	Refs	Part	Band faces
☐	D17–D24	1N914	North
☐	D29	1N914	East

Step 9 Reset and Break-key diodes

☐	Refs	Part	Band faces
☐	D25, D26, D31, D32	1N914	West
☐	D27	1N914	North
☐	D30	1N914	East

Step 10 Cassette, RS-232 and power diodes

☐	Refs	Part	Band faces
☐	D9	1N914	East
☐	D10	1N914	West
☐	D16	1N914 (short 5 mm spacing)	North
☐	D15	1N4001	South

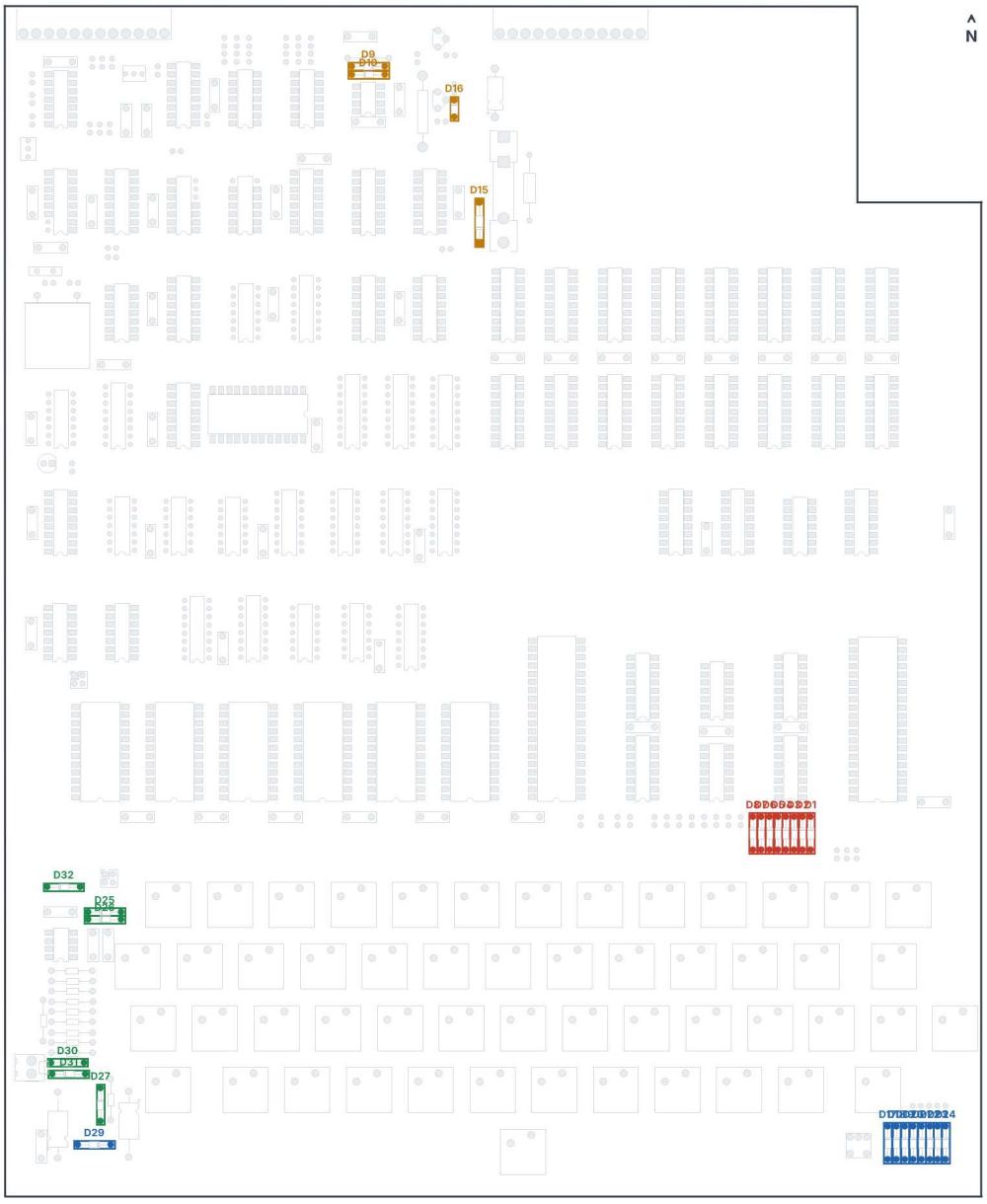


Figure 2-3 Steps 7–10: keyboard (red), DAC (blue), reset (green), cassette, RS-232 and power (orange) diodes.

Part 3 Resistors

Most resistors stand upright in holes 2.54 mm apart: bend one lead back over the body. The ones marked “flat” lie on the board in holes 12.7 mm apart (R64 and R65 are longer ½ W parts). Resistors are not polarised.

Step 11 Keyboard pull-ups

☐	Refs	Value	Mounting
☐	R1–R8	4.7 kΩ	Upright

Step 12 Sound DAC

☐	Ref	Value	Mounting
☐	R67	8.2 kΩ	Upright
☐	R68	3.9 kΩ	Upright
☐	R69	2 kΩ	Upright
☐	R70	16 kΩ	Upright
☐	R71	33 kΩ	Upright
☐	R73	68 kΩ	Upright
☐	R74	510 Ω	Upright
☐	R75	1 kΩ	Upright
☐	R89	510 Ω	Flat

Step 13 Reset and Break key

☐	Ref	Value	Mounting
☐	R76	4.7 kΩ	Flat
☐	R77	1 MΩ	Flat
☐	R78	100 kΩ	Flat
☐	R79	100 kΩ	Flat
☐	R80	1 MΩ	Flat
☐	R81	1 MΩ	Flat
☐	R82	1 MΩ	Flat
☐	R83	10 kΩ	Flat
☐	R84	100 kΩ	Flat
☐	R85	10 MΩ	Flat

Step 14 Processor and bus

☐	Ref	Value	Mounting
☐	R9	220 Ω	Upright
☐	R10, R11	4.7 kΩ	Upright
–	R88	390 Ω	Do not fit

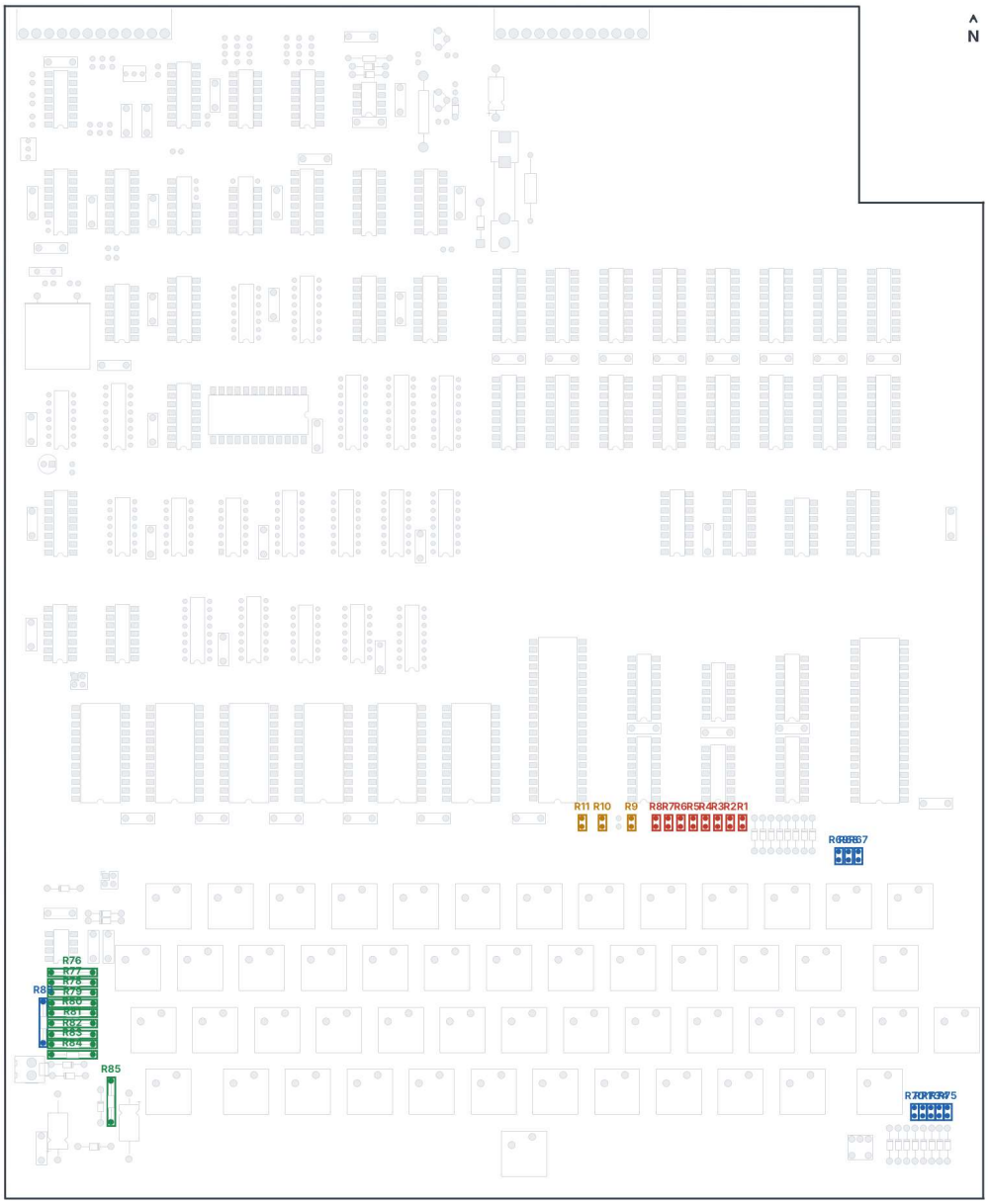


Figure 2-4 Steps 11–14: keyboard (red), DAC (blue), reset (green), processor (orange).

Step 15 Oscillator and video

☐	Ref	Value	Mounting
☐	R32	1 kΩ	Upright
☐	R33	10 kΩ	Upright
☐	R34	10 kΩ	Upright
☐	R35	470 Ω	Upright
☐	R36	470 Ω	Upright
☐	R50	10 kΩ	Upright
☐	R51	220 Ω	Upright
☐	R52	1 kΩ	Upright
☐	R59	1 kΩ	Upright
☐	R60	470 Ω	Upright
☐	R61	220 Ω	Upright
☐	R86	1 kΩ	Upright

Step 16 Cassette, RS-232 and LED

☐	Ref	Value	Mounting
☐	R37	10 kΩ	Flat
☐	R53	10 kΩ	Upright
☐	R54	1 kΩ	Upright
☐	R55	10 kΩ	Upright
☐	R56	100 kΩ	Upright
☐	R90	100 Ω	Upright
☐	R62	4.7 kΩ	Upright
☐	R63	10 kΩ	Upright
☐	R66	10 kΩ	Upright
☐	R72	1 kΩ	Upright
☐	R64	1 kΩ ½ W	Flat, long
☐	R65	1 kΩ ½ W	Flat, long
☐	R87	470 Ω	Upright

Note R38–R49 are the termination resistors for the external serial port on J2. They were normally not installed; leave them out unless you are fitting U67 and U68.

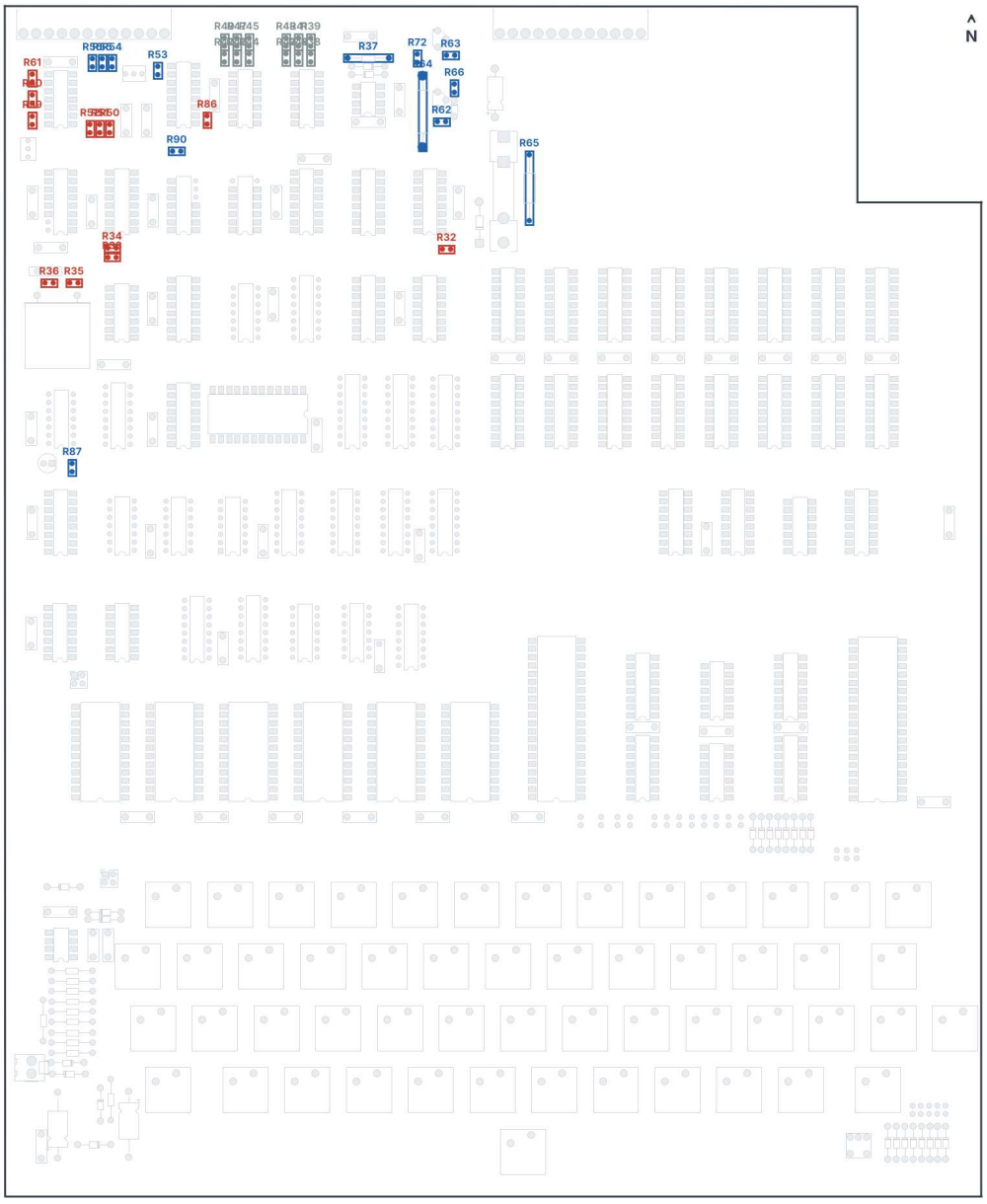


Figure 2-5 Steps 15 and 16: oscillator and video (red), cassette, RS-232 and LED (blue). Grey: optional R38–R49.

Part 4 Capacitors

Step 17 Decoupling capacitors

39 capacitors of 0.1 µF (marked 104), one beside each group of chips.

☐	Refs	Value	Count
☐	C21–C30, C32, C33, C35–C54, C56, C58, C59, C66–C69	0.1 µF	39

Step 18 Timing and signal capacitors

☐	Ref	Value	Marking
☐	C6	150 pF	151
☐	C7	0.001 µF	102
☐	C8	0.1 µF	104
☐	C9	68 pF	68 or 680
☐	C10	0.01 µF	103
☐	C11	0.1 µF	104
☐	C12	0.001 µF	102
☐	C13	0.01 µF	103
☐	C55	0.001 µF	102
☐	C57	27 pF (disc)	27 or 270
☐	C65	150 pF	151

Step 19 Reset capacitors

☐	Ref	Value	Marking
☐	C60	0.001 µF	102
☐	C61	150 pF	151
☐	C62	0.1 µF	104

Step 20 Electrolytic capacitors

These are polarised. Fit them lying flat, + lead in the square pad, rated at 16 V or more.

☐	Ref	Value	+ lead faces
☐	C5	47 µF	South
☐	C63	33–35 µF	South
☐	C64	47–50 µF	North

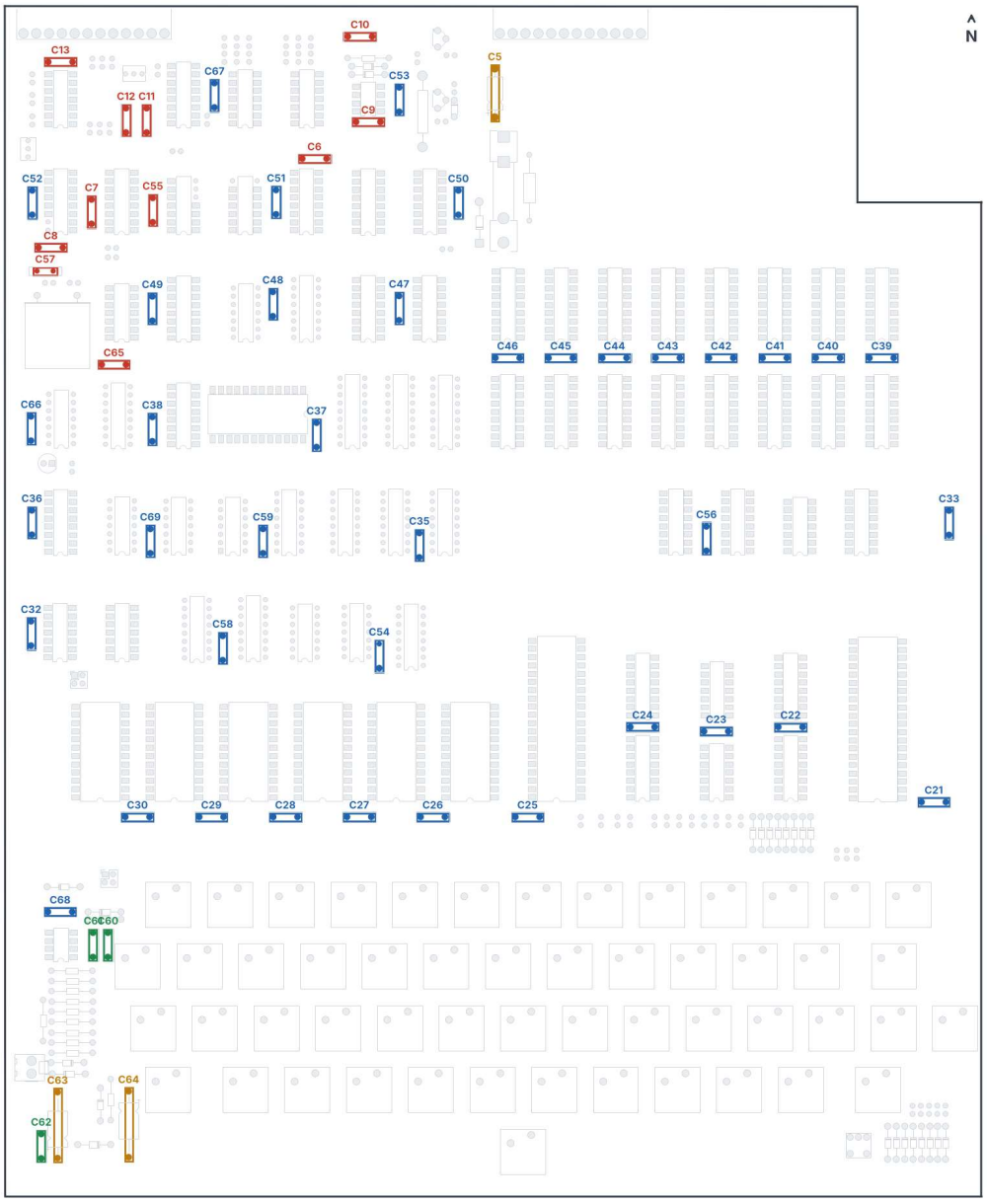


Figure 2-6 Steps 17–20: decoupling (blue), timing (red), reset (green), electrolytic (orange).

Part 5 Other components

Step 21 Trimmers and transistors

☐	Ref	Part	Notes
☐	R57	10 kΩ trimmer (Bourns 3266Y or similar)	Cassette input level
☐	R58	5 kΩ trimmer	Video level
☐	Q1	2N5226 PNP	Flat face to the silkscreen. Pads: 1 collector, 2 base, 3 emitter
☐	Q2	2N5225 NPN	Flat face to the silkscreen. Pads: 1 collector, 2 base, 3 emitter

> **Note** 2N5225 and 2N5226 are long obsolete. 2N3904 (NPN) and 2N3906 (PNP) work in this circuit, but their legs are in a different order. Bend the leads so collector, base and emitter land in pads 1, 2 and 3.

Step 22 LED and crystal

☐	Ref	Part	Notes
☐	LED1	5 mm red LED	Flat side (cathode) East, in the square pad
☐	X1	3.93216 MHz crystal (HC-51 can)	Lay it flat on its silkscreen outline. Avoid heating the can.

Step 23 Connectors, headers and fuse clips

☐	Ref	Part	Notes
☐	J2	12-way right-angle Molex header	Video, cassette and external serial
☐	J3	12-way right-angle Molex header	Serial options and baud rate (see Section 8)
☐	J4	2-way Molex KK-396 header	Speaker output from the sound DAC
☐	W3, W5	2×2 pin headers	ACIA CTS and receive-clock options
☐	F1	Fuse clips for 6.3 × 32 mm fuse	Do not fit the fuse yet

Step 24 Power leads and feet

- Solder a red lead to the +5V_IN holes and a black lead to the ground holes beside the fuse clips in the north-east corner. Use wire rated for at least 3 A, and add a strain relief.
- Fit rubber feet or standoffs in the mounting holes. Keep the two holes either side of the space bar free for the space-bar fitting.

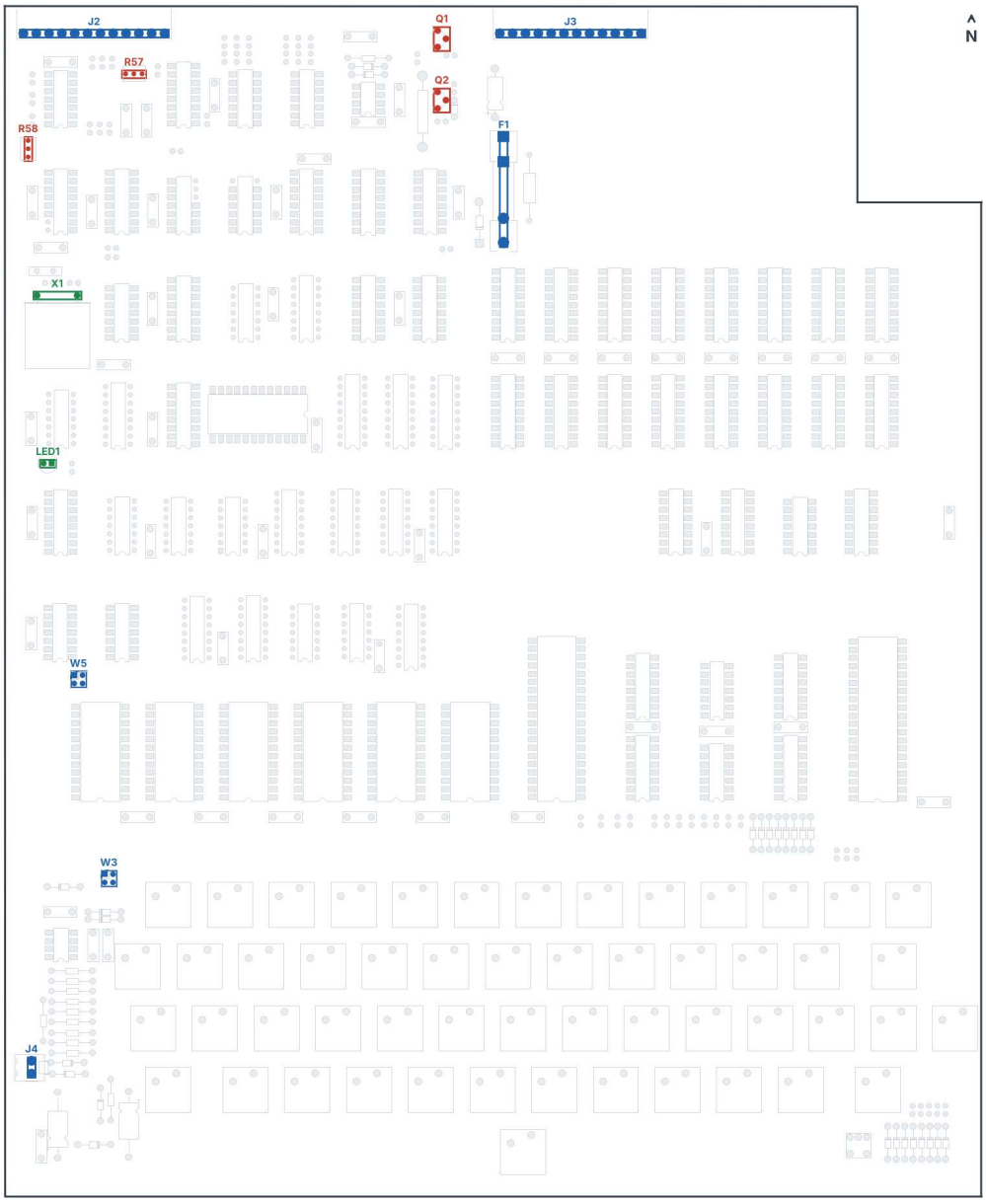


Figure 2-7 Steps 21–23: trimmers and transistors (red), LED and crystal (green), connectors and fuse clips (blue).

Note This completes Stage I. Clean off flux residue with isopropyl alcohol and check every joint under a good light before you start the keyboard.

3 Stage II – keyboard

The keyboard uses PCB-mount Cherry MX keyswitches. The five-pin type, with two extra plastic pegs, locates itself in the board; three-pin plate-mount switches also fit but need more care to keep straight.

Key positions

Table 3-1 lists every switch position in keyboard order, from the top row to the space bar. Use it to place your keycaps.

Row	Switch and key
Row 1 (top)	SW1 1, SW2 2, SW3 3, SW4 4, SW5 5, SW6 6, SW7 7, SW8 8, SW9 9, SW10 0, SW11 :, SW12 -, SW13 RUB OUT
Row 2	SW14 ESC, SW15 Q, SW16 W, SW17 E, SW18 R, SW19 T, SW20 Y, SW21 U, SW22 I, SW23 O, SW24 P, SW25 LINE FEED, SW26 RETURN
Row 3 (home)	SW27 CTRL, SW28 A, SW29 S, SW30 D, SW31 F, SW32 G, SW33 H, SW34 J, SW35 K, SW36 L, SW37 ;, SW38 SHIFT LOCK, SW39 REPEAT, SW40 BREAK
Row 4	SW41 L SHIFT, SW42 Z, SW43 X, SW44 C, SW45 V, SW46 B, SW47 N, SW48 M, SW49 ,, SW50 ., SW51 /, SW52 R SHIFT
Space bar	SW53 SPACE
Shift lock (alternative)	SW54 SHIFT LOCK

RETURN (SW26) and the two SHIFT keys (SW41, SW52) are 1.5-unit positions; all the others are 1 unit wide.

Step 25 Prepare the special keys

- Shift lock: choose one position. Either fit a latching Cherry MX Lock switch at SW38 (in the home row, as on the original OSI layout), or fit a latching switch at SW54 in the south-east corner. Never fit both, because they are wired in parallel.
- Break key SW40: snip off the right-hand (east) plastic locating peg, flush with the switch base, before fitting. That hole was removed from the board because it cut into the ground strip. The spot is marked SNIP PEG on the silkscreen.
- Space bar SW53: the board has no Cherry stabiliser footprint. The two holes either side of the space bar are the original OSI space-bar mounting holes. Decide how you will support the space bar before soldering SW53.

Step 26 Fit and solder the switches

- Work one row at a time, starting with the top row.
- Push each switch fully home so its base sits flat on the board, with all pegs in their holes.
- Solder one pin of every switch in the row, check that they are all square and flat, then solder the second pin. Re-melt a joint and press the switch down if one has lifted.
- Do not hold the iron on a switch pin for more than a few seconds; the plastic housing softens.
- Fit SW40 (Break) after the home row, with its peg snipped.

Step 27 Space bar and keycaps

- Fit SW53 and your space-bar support.
- Fit the keycaps using Table 3-1. Check that every key moves freely and does not rub

on its neighbours.

Test 4 Keyboard matrix (optional, before chips are fitted)

With the meter on continuity, put the probes on the two pins of a switch from underneath and press the key: it should read closed only while the key is held. A quick pass over all 54 positions catches unsoldered pins.

> **Note** This completes all the soldering. The rest of the build is fitting chips and testing.

4 Stage III – power and video

In this stage you first check the power wiring with no chips fitted, then fit the video chips and get a picture on the screen. The video section works without the processor, so a screen full of characters proves a large part of the board.

Chip handling

- Pin 1 of each chip goes to the notched end of its socket: notch South everywhere, except U41 whose notch faces East.
- Straighten the legs by pressing each row gently against a flat surface before inserting.
- Start the pins at one end, check that they all line up, then press the chip home. Support the board underneath while you push.
- After fitting, look along each side of every chip for legs that have folded under.
- NMOS and CMOS chips (6502, 6850, 2114, EPROMs, CA3130) are static sensitive: touch a grounded surface first and handle them by the body.

Step 28 Power-up with no chips

- With no chips fitted and the supply disconnected, measure the resistance between +5V_IN and ground. It must not read as a short (under about 10 Ω); a few hundred ohms is normal, because of the resistors connected across the supply.
- Measure between the ground holes and pin 18 of U31 (that is +5 V on the board side of the fuse). Again it must not be a short.
- Fit a 5 A fuse in F1.
- Connect the red lead to +5 V and the black lead to the supply's negative terminal.

Test 5 Power supply

- Switch on briefly. LED1 should light. If it does not, switch off at once and check the supply polarity.
- Measure the DC voltage between U8 pin 8 (+5 V) and U8 pin 21 (ground). It should be between 4.8 V and 5.2 V. More than 5.2 V can damage the chips.
- Set the meter to AC and measure across C5. The ripple must not be more than 0.2 V.
- Spot-check +5 V on pin 24 of the ROM sockets, pin 18 of the 2114 sockets, and pin 14 or 16 of a few logic sockets.
- Switch off.

Step 29 Fit the video chips

☐	Refs	Chip
☐	U58	74LS00
☐	U70	74LS03
☐	U43	74LS08
☐	U65, U69	74LS123
☐	U63, U76	74LS74
☐	U62	74LS04
☐	U53, U54, U55, U78	74LS157
☐	U30, U57, U59, U60, U61	74LS163
☐	U42	74LS165
☐	U19, U56	74LS20

☐	U64	74LS76
☐	U79	74LS86
☐	U29	74LS93
☐	U24, U25	8T28
☐	U39, U40	2114 (video RAM)
☐	U41	Character generator 2716 – notch East
☐	U66	CA3130
☐	U80	LM393

Note Fit U74 (8T28) and U77 (2114) as well if you are using the colour option on J75.

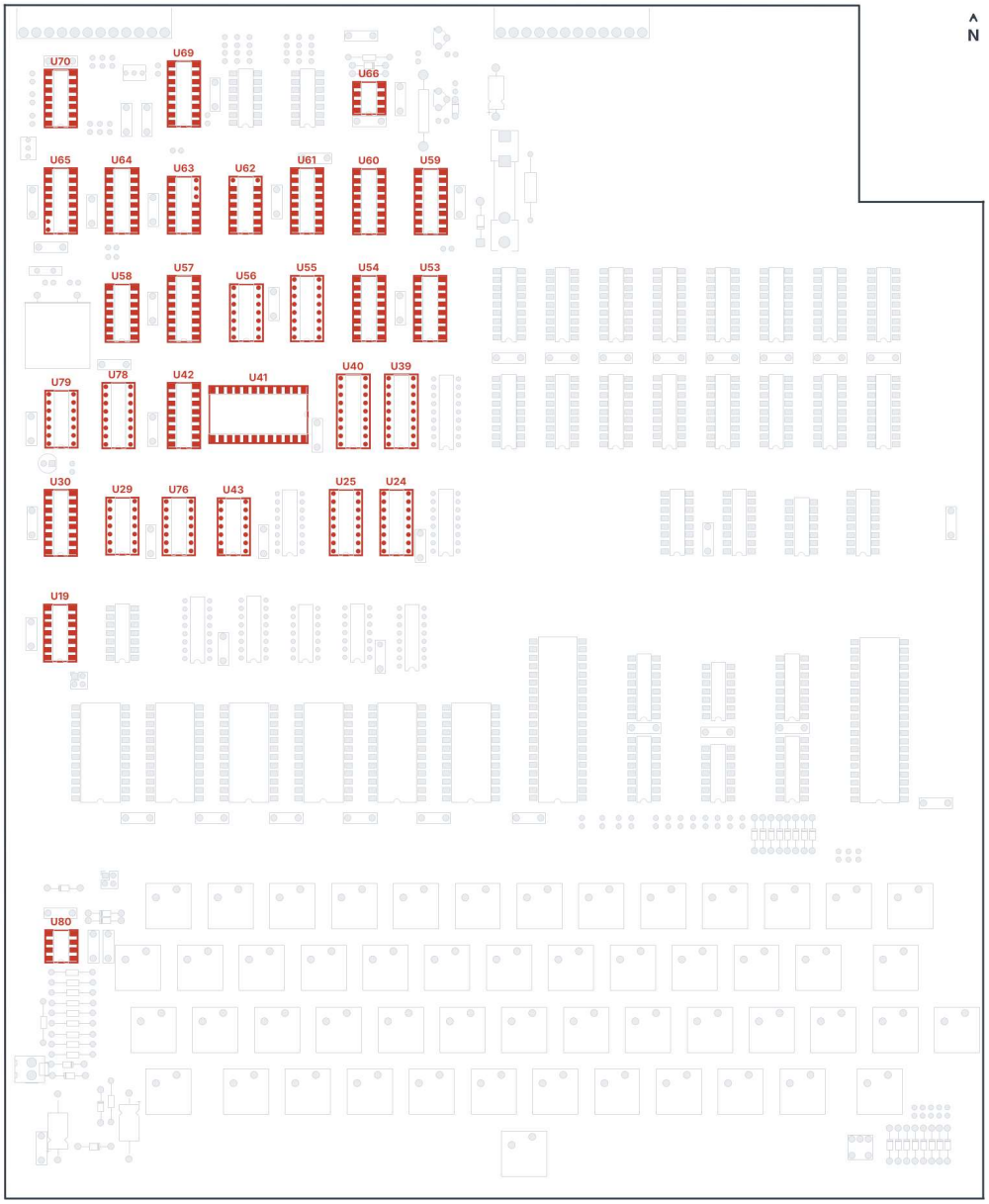


Figure 4-1 Step 29: video, timing and cassette chips.

Step 30 Connect the monitor

J2 carries video and cassette signals:

J2 pin	Signal	Connect to
12	Video out	Monitor video input (centre)
11	Ground	Monitor video screen
10	Tape in (MON)	Recorder earphone or line output
9	Aux out	Optional
8	Ground	Recorder ground
7	Tape out (MIC)	Recorder microphone input

Test 6 Video

- Turn on the monitor, then the board.
- Adjust R58 until the picture is steady and clear. You should see a screen full of the same graphics character, or a random mixture of characters – either is correct at this stage.
- Set R57 fully clockwise, then back a quarter turn. This sets the cassette input and can be fine-tuned later.
- With a logic probe you can check for pulses on J2 pin 12 (video), U60 pin 14, U30 pins 11–15 (video counters), U8 pin 37 (CPU clock, present even with no CPU) and U58 pin 8 (the 3.93 MHz oscillator; some probes are too slow to see it).
- Switch off.

5 Stage IV – processor, ROMs and RAM

Disconnect the power before each step. All chips in this stage fit with the notch to the South.

Part 1 Processor, decoding, monitor ROM and the first 1K of RAM

Step 31

☐	Refs	Chip
☐	U8	6502 processor
☐	U4, U5	74LS125
☐	U2, U3	74LS75
☐	U20, U22, U23, U73	74LS138
☐	U16, U18, U21	74LS04
☐	U15	74LS02
☐	U17	74LS139
☐	U72	74LS174
☐	U13	Monitor ROM (SYN600 image)
☐	U12	BASIC 4 ROM
☐	U31, U45	2114 – the first 1K of RAM at $0000

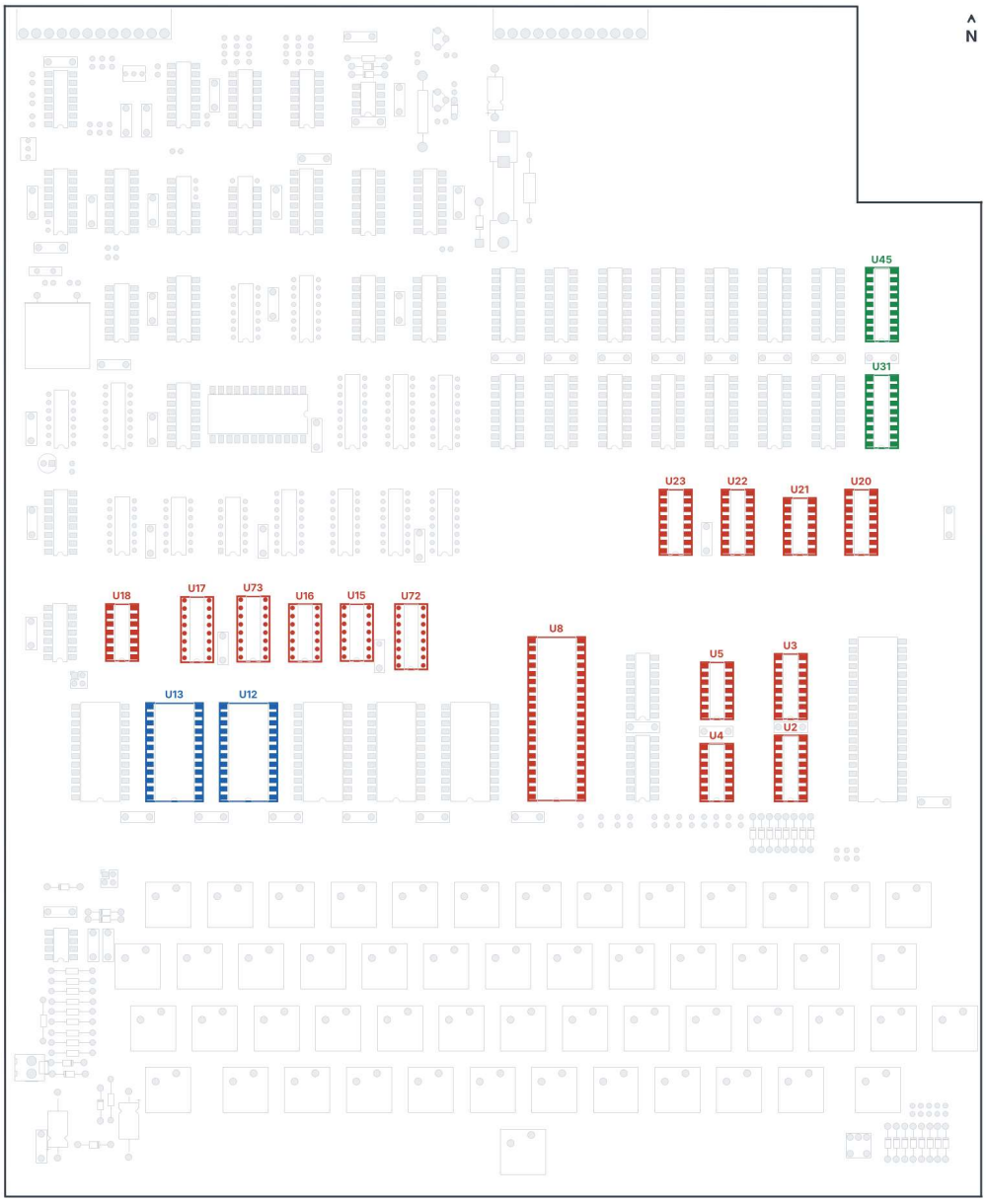


Figure 5-1 Step 31: processor and logic (red), monitor and BASIC 4 ROMs (blue), first 1K of RAM (green).

Note BASIC 4 goes in with the monitor in this step, as in the original build order.

Test 7 Monitor

- Apply power with the monitor connected. D/C/W/M ? should appear at the bottom left of the screen. If it does not, see Section 10.
- Make sure SHIFT LOCK is down. At this point only the M and BREAK keys do anything.
- Press M. The screen shows 0000 followed by two hexadecimal digits: the address and the contents of the first memory location.
- Type an address with the number keys and A–F. Press / to change the contents, type a value, and press RETURN to step to the next address. Press . to enter a new address.
- Try writing and reading back a few values in the first 1K ($0000–$03FF). At $F800 the monitor ROM reads A0 and cannot be changed.

- If nothing happens, check SHIFT LOCK, hold BREAK for about two seconds, release it and press M again.

Part 2 ACIA and BASIC ROMs

Step 32

☐	Refs	Chip
☐	U14	6850 ACIA
☐	U11	BASIC 3 ROM
☐	U10	BASIC 2 ROM
☐	U9	BASIC 1 ROM

Test 8 BASIC

- Apply power. At D/C/W/M ? (SHIFT LOCK down) press C. MEMORY SIZE? appears.
- Press RETURN. TERMINAL WIDTH? appears. Press RETURN again.
- The screen should show 255 BYTES FREE, the OSI 6502 BASIC version message and OK.
- Hold BREAK until D/C/W/M ? appears, then press W (warm start). The OK prompt should return.

Part 3 The remaining RAM

Step 33

☐	Refs	Chip
☐	U32–U38, U46–U52	2114 RAM, 14 chips

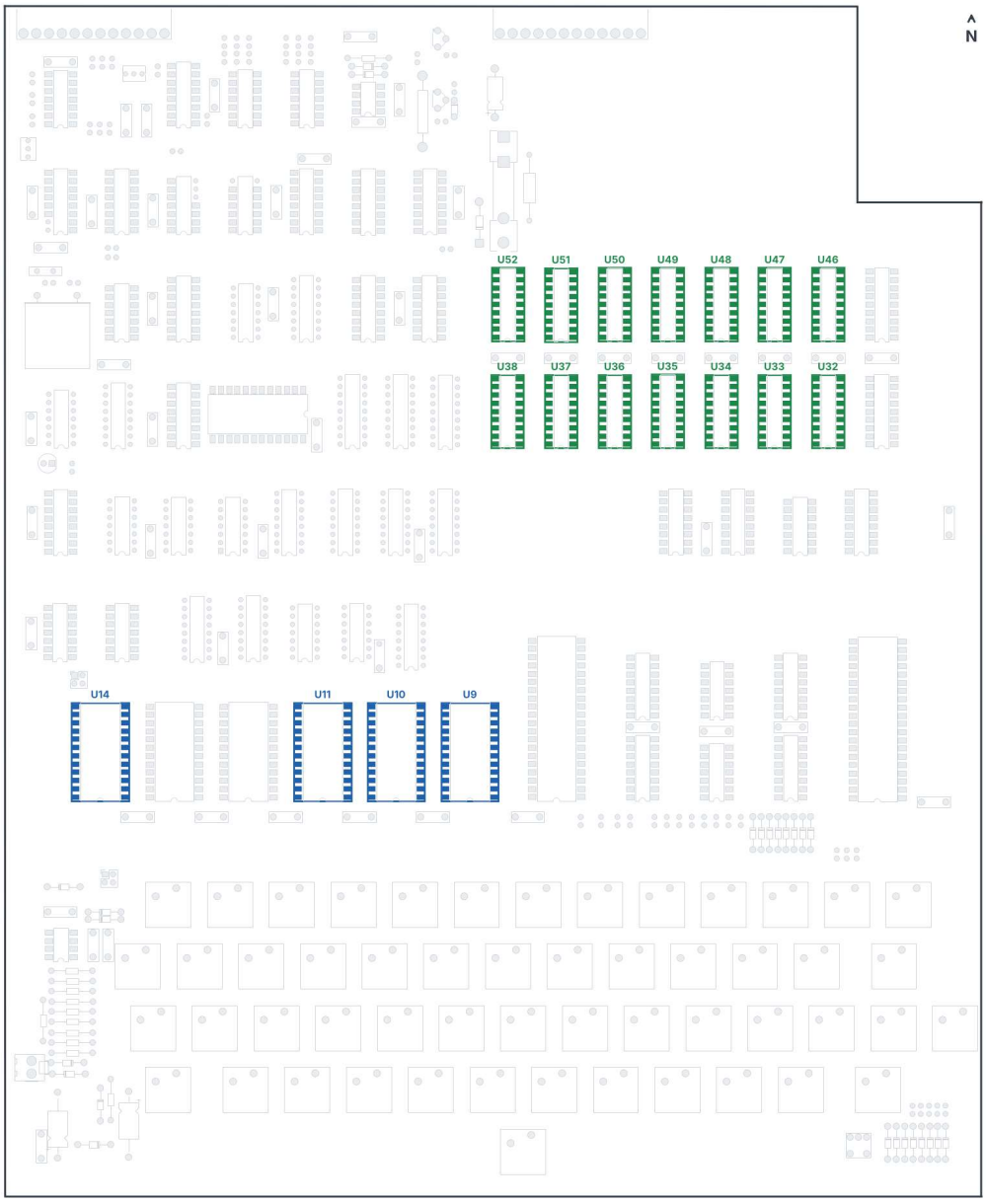


Figure 5-2 Steps 32 and 33: ACIA and BASIC 1–3 (blue), remaining RAM (green).

Test 9 Memory size

- Cold start with C and press RETURN twice.
- The screen should show 7423 BYTES FREE. Any other number points to a RAM chip or the RAM select logic (U22, U23).

Done Congratulations – your Superboard II is complete. It works the same as an Ohio Scientific Challenger 1P, so the original user manuals and software apply.

6 Optional parts

Parts	Purpose	Fit when
U6, U7 (8T28)	Buffers the data bus to the J1 expansion connector	You attach a 610 or other expansion board to J1
U74 (8T28), U77 (2114), J75	Colour RAM and colour connector	You use a colour board on J75
U67 (74LS14), U68 (74LS17), R38–R49	External serial port on J2 pins 1–6	You use the external serial lines
R88 (390 Ω)	DD line pull-down	Normally not fitted

7 Jumper settings

W3 – ACIA CTS

The 6850 ACIA will only transmit when its CTS input is low. Fit a link across W3 pins 1–2 (or 3–4) to tie CTS to ground. This is needed for saving to cassette and for serial output. The same job can be done on J3 (pins 7 and 9).

W5 – receive clock

Leave W5 open: the ACIA transmit and receive clocks are joined on the board and driven by the baud-rate divider. Link pins 1–2 only if you fit the optional external-serial parts and want the receive clock from J2 pin 2.

8 J3 option connector

J3 selects the serial data source and the baud rate, using a plug with wire links (the original kit used a shorting plug).

J3 pin	Signal	J3 pin	Signal
1	Ground	7	Ground (–V IN, tied to ground on this board)
2	RS-232 out	8	Baud divider input
3	RS-232 in	9	ACIA CTS
4	ACIA receive data	10	124.8 kHz clock
5	Cassette receive data	11	+5 V
6	RS-232 receive data	12	499.2 kHz clock

Setting	Links
Cassette at 300 baud (standard)	4–5, 8–10 and 7–9 (or use W3 for CTS)
Cassette or serial at 1200 baud	Replace 8–10 with 8–12
RS-232 serial	4–6 instead of 4–5; signals on pins 2 and 3

Note Because –V IN is tied to ground, the RS-232 output only swings between 0 and +5 V. Most modern USB serial adapters accept this, but a true ±12 V device may not.

9 Programming the ROMs

Socket	Contents	Address	Part
U13	Monitor (SYN600)	$F800–$FFFF	27C16 or 2816
U9	BASIC 1	$A000–$A7FF	27C16 or 2816
U10	BASIC 2	$A800–$AFFF	27C16 or 2816
U11	BASIC 3	$B000–$B7FF	27C16 or 2816
U12	BASIC 4	$B800–$BFFF	27C16 or 2816
U41	Character generator	Video only	2716 or 27C16

- Each image is 2 KB. Use the Superboard II / Challenger 1P images; images from other OSI machines will not start correctly.
- Use parts with an access time of 450 ns or faster.
- A 2816 EEPROM works in the program sockets with no change; its write-enable pin (21) is tied to +5 V, so the board can only read it.
- The character generator image must be the OSI 600 character set in 2716 layout, as on the original character ROM.

10 Troubleshooting

Symptom	Likely causes
Blank screen at power-up	Video cable; R58 setting; crystal oscillator (X1, U58); video timing (U29, U30, U59–U61); monitor ROM U13; BASIC 4 (U12)
Random characters, or BREAK does nothing	Reset circuit: U80, R76–R85, C60–C62, C64, D25–D27, D30–D32
Picture rolls	Vertical sync one-shot U65 and its parts; timing counter U61
Split screen (2 or 4 images)	Video timing U59; video address multiplexers U53–U55
Wrong number of bytes free	A faulty 2114, or the RAM selects U22 and U23
No OK prompt after cold start	BASIC 1 (U9)
Goes to the monitor but will not cold start	A BASIC ROM; ROM select U73; the first 1K of RAM (U31, U45)
No lower-case characters	Character generator U41
One key or a group of keys dead	That switch's solder joints; keyboard decoding U2–U5; diodes D1–D8
Cassette will not load	R57 setting; recorder volume; J3 links; U66 and U62
Cassette will not save, no serial output	CTS not grounded (W3 or J3 7–9); U14

Appendix A Bill of materials

Ref	Value	Qty	Refs
IC	6502	1	U8
IC	6850	1	U14
IC	27C16 or 2816 (programmed)	5	U9–U13
IC	2716 character generator	1	U41
IC	2114 RAM	19	U31–U40, U45–U52, U77*
IC	8T28	5	U24, U25, U6*, U7*, U74*
IC	74LS00	1	U58
IC	74LS02	1	U15
IC	74LS03	1	U70
IC	74LS04	4	U16, U18, U21, U62
IC	74LS08	1	U43
IC	74LS14	1	U67*
IC	74LS17	1	U68*
IC	74LS20	2	U19, U56
IC	74LS74	2	U63, U76
IC	74LS75	2	U2, U3
IC	74LS76	1	U64
IC	74LS86	1	U79
IC	74LS93	1	U29
IC	74LS123	2	U65, U69
IC	74LS125	2	U4, U5
IC	74LS138	4	U20, U22, U23, U73
IC	74LS139	1	U17
IC	74LS157	4	U53, U54, U55, U78
IC	74LS163	5	U30, U57, U59, U60, U61
IC	74LS165	1	U42
IC	74LS174	1	U72
IC	CA3130	1	U66
IC	LM393	1	U80
Socket	40 / 24 / 18 / 16 / 14 / 8 pin	2 / 7 / 19 / 27 / 18 / 2	See Stage I, Part 1
Diode	1N914 or 1N4148	26	D1–D10, D16–D27, D29–D32
Diode	1N4001	1	D15
LED	5 mm red	1	LED1
Transistor	2N5225 NPN	1	Q2
Transistor	2N5226 PNP	1	Q1
Crystal	3.93216 MHz HC-51	1	X1
Resistor	100 Ω	1	R90
Resistor	220 Ω	3 (+6*)	R9, R51, R61 (R38, R40, R42, R44, R46, R48*)
Resistor	390 Ω	0 (+7*)	R39, R41, R43, R45, R47, R49*, R88 (not fitted)
Resistor	470 Ω	4	R35, R36, R60, R87
Resistor	510 Ω	2	R74, R89
Resistor	1 kΩ	7	R32, R52, R54, R59, R72, R75, R86
Resistor	1 kΩ ½ W	2	R64, R65
Resistor	2 kΩ	1	R69
Resistor	3.9 kΩ	1	R68
Resistor	4.7 kΩ	12	R1–R8, R10, R11, R62, R76
Resistor	8.2 kΩ	1	R67
Resistor	10 kΩ	9	R33, R34, R37, R50, R53, R55, R63, R66, R83
Resistor	16 kΩ	1	R70
Resistor	33 kΩ	1	R71
Resistor	68 kΩ	1	R73
Resistor	100 kΩ	4	R56, R78, R79, R84
Resistor	1 MΩ	4	R77, R80, R81, R82
Resistor	10 MΩ	1	R85
Trimmer	10 kΩ (3266Y)	1	R57
Trimmer	5 kΩ (3266Y)	1	R58

Capacitor	0.1 µF	42	C8, C11, C62 and 39 decoupling (Step 17)
Capacitor	0.01 µF	2	C10, C13
Capacitor	0.001 µF	4	C7, C12, C55, C60
Capacitor	150 pF	3	C6, C61, C65
Capacitor	68 pF	1	C9
Capacitor	27 pF disc	1	C57
Electrolytic	47 µF 16 V	1	C5
Electrolytic	33–35 µF 16 V	1	C63
Electrolytic	47–50 µF 16 V	1	C64
Connector	12-way right-angle Molex	2	J2, J3
Connector	2-way Molex KK-396	1	J4
Header	2×2 pin header and link	2	W3, W5
Fuse	Clips for 6.3 × 32 mm, 5 A fuse	1	F1
Switch	Cherry MX PCB-mount	53	SW1–SW53
Switch	Latching (shift lock)	1	SW38 (MX Lock) or SW54
Keycaps	OSI layout (Table 3-1)	54	

* Optional parts, see Section 6.

Appendix B Memory map

Address	Contents
$0000–$1FFF	8K program RAM (U31–U38, U45–U52; 1K per pair)
$A000–$BFFF	BASIC ROMs U9–U12
$D000–$D3FF	Video RAM (U39, U40)
$D400–$D7FF	Colour RAM (U77, optional)
$D800	Control register U72 (32/64 characters, colour, sound DAC)
$DC00	Keyboard (write row, read column)
$F000	ACIA 6850
$F800–$FFFF	Monitor ROM U13

Appendix C Connector pinouts

J2 pin	Signal	J2 pin	Signal
1	External Rx data*	7	Tape out (MIC)
2	External Rx clock*	8	Ground
3	CTS*	9	Aux out
4	External Tx data*	10	Tape in (MON)
5	External Tx clock*	11	Ground
6	RTS*	12	Video out

* Uses the optional U67, U68 and R38–R49. J4: pin 1 ground, pin 2 sound output. J3: see Section 8.

Appendix D Original manual references

The original Ohio Scientific Superboard Kit Assembly Manual (April 1982) remains useful for its character graphics chart, BASIC error codes and operating notes. Its step numbers, part bags and test pin numbers do not match this board.

Appendix E Reference changes from the OSI schematics

On the OSI sheets	On this board	Part
R74 (sheet 1)	R88	390 Ω DD pull-down, not fitted

R62A (sheet 3)	R90	100 Ω cassette input
R86 (sheet 5)	R89	510 Ω sound DAC pull-up; R86 on sheet 3 keeps its number
D3 LED (sheet 5)	LED1	Power LED
R73 470 Ω (sheet 5)	R87	LED resistor
Shift lock	SW54 or SW38	Shift-lock switch
Keys	SW1–SW53	Cherry MX switches; SW40 is Break